Table R1: GCB component ablations and pruning comparison on Gwilliams zero-shot. *Component ablations*: w/o GCB is the no-aggregation baseline; w/o gate removes the quantile threshold and uses raw top-K scores, dropping R@1 below baseline and confirming that selective aggregation of reliable evidence is necessary; w/o norm† removes bucket-size normalisation, inflating R@1 via sentence-length bias (Spearman $r{=}0.27$, $p{<}0.001$ between bucket size and gain), and is therefore not directly comparable. *Pruning comparison*: hard pruning restricts retrieval to the top-S buckets by setting all remaining candidate logits to $-\infty$, physically removing them from ranking. Despite achieving higher R@10 by concentrating recall within fewer buckets, hard pruning underperforms GCB-full on the top-rank metrics R@1 and MRR, showing that the GCB gain is not explained by implicit candidate-pool reduction.

Method	R@1	R@5	R@10	MRR	MedR
Component ablations					
GCB-full	**0.527**	**0.714**	0.777	**0.614**	1
w/o gate	0.378	0.640	0.728	0.498	3
w/o norm†	0.712	0.856	0.885	0.779	1
w/o GCB	0.441	0.657	0.736	0.542	2
Pruning comparison ($S{=}3$)					
Hard pruning	0.505	0.623	**0.823**	0.582	1

Table R2: Multi-seed GCB Full vs. Baseline across datasets and backbones. Point estimates are mean query-level metrics across runs. Parentheses show 95% paired cluster-bootstrap confidence interval for the across-run mean GCB improvement, using sentence as the resampling unit; the interval is reported on the absolute GCB metric scale. $^*p{<}0.05$ for the across-run mean improvement. R@10 and MedR are point estimates only. **Bold: best backbone per metric within each setting.** Zero-shot: 3 seeds, consistent with the single-seed values in the main submission, as the zero-shot split is fixed by content identity and unaffected by seed; session-isolation: 5folds $\times$ 3seeds, values differ from the main submission because this setting constructs subject/session-level folds per seed, so the test population itself varies across seeds; the backbone ordering and direction of GCB gains remain consistent across all configurations.

Backbone	Baseline				+GCB Full			
	R@1	MRR	R@10	MedR	R@1	MRR	R@10	MedR
Gwilliams, zero-shot decoding ($N{=}1{,}464$ candidate windows)								
Dense CNN	**0.444**	**0.544**	**0.736**	**2.0**	**0.521 (0.516–0.527)***	**0.608 (0.604–0.613)***	**0.773**	**1.0**
Exp-dilated ($d{=}5$)	0.419	0.519	0.711	**2.0**	0.506 (0.501–0.512)*	0.592 (0.588–0.597)*	0.754	1.3
Exp-dilated ($d{=}10$)	0.396	0.497	0.692	3.0	0.462 (0.457–0.468)*	0.553 (0.549–0.558)*	0.727	2.0
Gwilliams, session-isolation ($N{=}7{,}011$ candidate windows, 5 folds $\times$ 3 seeds)								
Dense CNN	**0.195**	**0.265**	**0.401**	**28.2**	**0.202 (0.200–0.204)***	**0.273 (0.270–0.275)***	**0.408**	**27.1**
Exp-dilated ($d{=}5$)	0.162	0.230	0.361	38.1	0.167 (0.167–0.168)*	0.236 (0.235–0.236)*	0.368	37.0
Exp-dilated ($d{=}10$)	0.151	0.217	0.345	42.9	0.154 (0.154–0.155)*	0.221 (0.221–0.222)*	0.351	42.0
MOUS, zero-shot decoding ($N{=}825$ candidate windows)								
Dense CNN	**0.217**	**0.302**	**0.466**	**13.3**	**0.286 (0.283–0.289)***	**0.370 (0.368–0.373)***	**0.528**	**8.3**
Exp-dilated ($d{=}5$)	0.161	0.240	0.395	21.7	0.215 (0.212–0.218)*	0.293 (0.290–0.295)*	0.440	16.7
Exp-dilated ($d{=}10$)	0.134	0.208	0.351	30.0	0.179 (0.176–0.182)*	0.252 (0.250–0.255)*	0.388	24.7

Table R3: Comparison of leakage-control protocol in Défossez et al. (2023) and our protocol. The two protocols share the same window formulation and candidate construction. Our sole additional control is a global stimulus-identity split: we define a content key as the tuple (audio file, onset, offset) and enforce that all recordings sharing the same key are assigned to the same partition, regardless of subject or session. This prevents repeated stimulus content from crossing train/test boundaries across subjects or sessions, which is not explicitly constrained in Défossez et al. (2023).

Mechanism	Défossez et al. (2023)	Our protocol
Window formulation	Fixed 3 s word-anchored windows	Identical
Candidate construction	Audio-window pool de-duplicated by stimulus identity and window index	Identical
Split assignment	Cross-subject/session consistency of repeated stimulus content is not explicitly specified as a global constraint	Global stimulus identity: tuple (audio file, onset, offset), enforced once across all subjects and sessions
Repeated content across subjects / sessions	Not explicitly controlled at the global stimulus-identity level	Prevented by construction: all recordings sharing the same stimulus-identity key are assigned to the same partition, regardless of subject or session

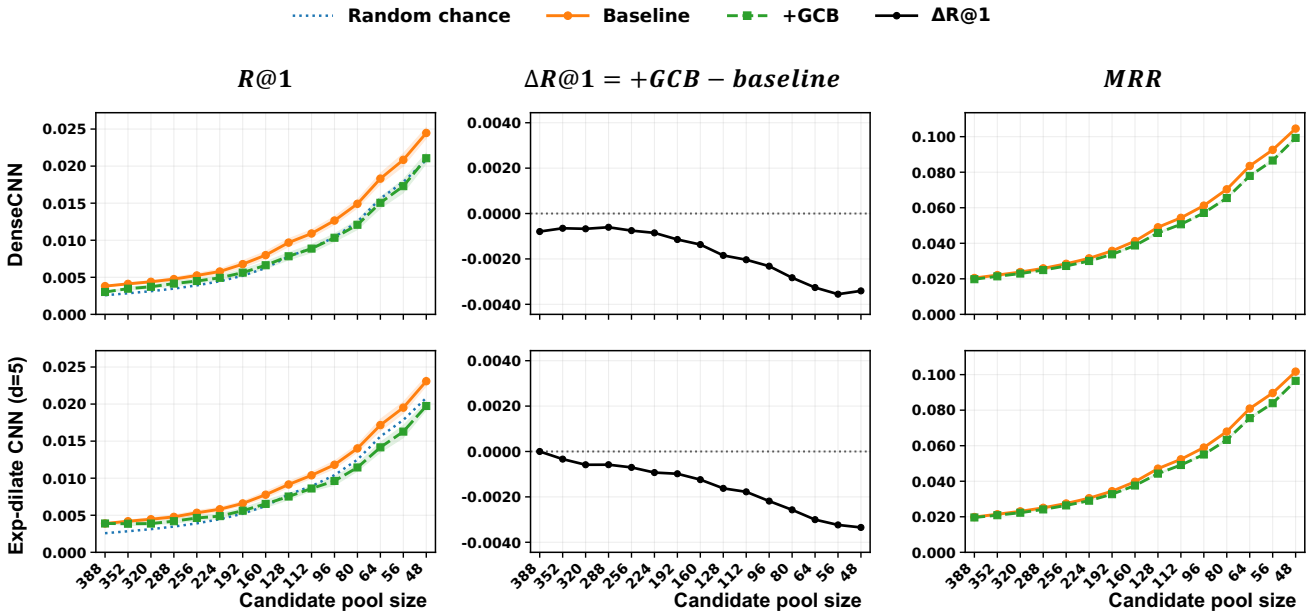

Figure R1: Brennan EEG retrieval performance across candidate pool sizes. As the candidate pool shrinks from $N{=}388$ to $N{=}48$, absolute R@1 and MRR improve monotonically for both backbones, confirming that the near-chance performance at the full pool size is partly driven by evaluation difficulty rather than a complete absence of decodable signal. However, ΔR@1 remains negative across all pool sizes for both backbones, meaning GCB consistently underperforms the no-aggregation baseline regardless of pool size. This pattern is consistent with an evidence-limited regime: when single-window neural evidence is too weak to produce reliable logit support, GCB aggregates noise rather than signal, and the bucket-wise bias introduces more errors than it corrects. The result serves as a leakage check: if GCB exploited sentence grouping structure rather than genuine neural evidence, it would be expected to produce gains even under weak signal, which is not observed here.

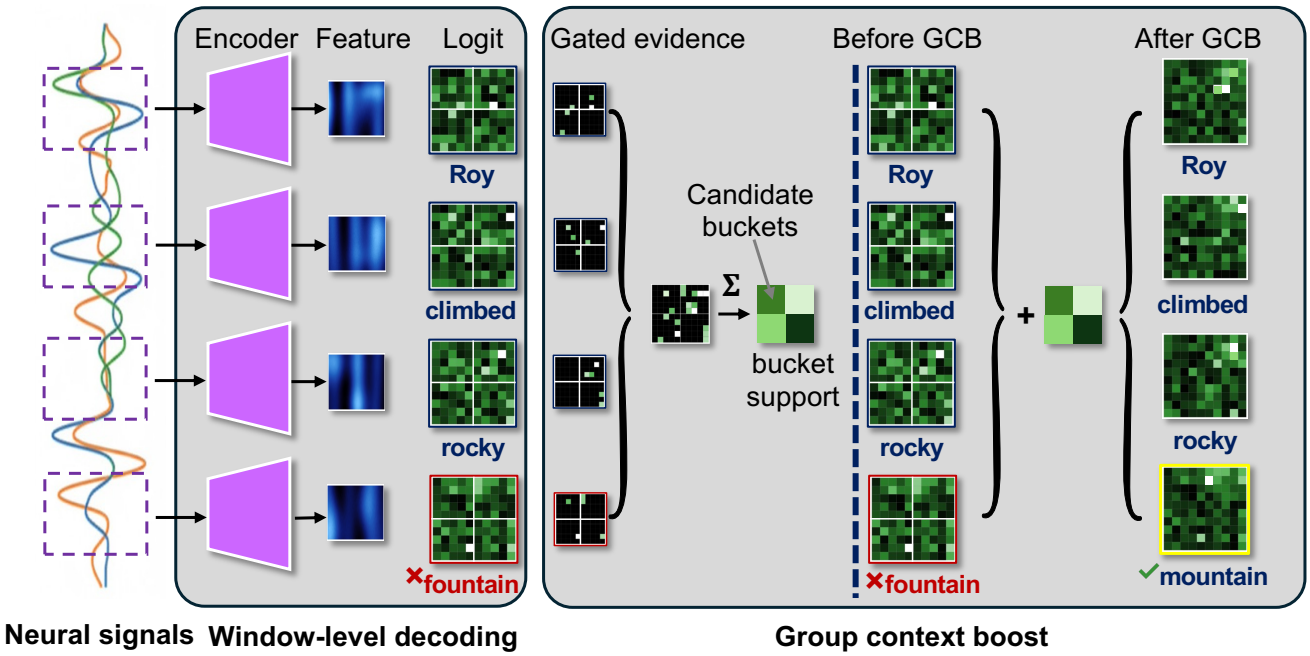

Figure R2: Simplified schematic of the GCB aggregation mechanism. Each neural window is encoded into logits over the candidate pool. GCB retains high-confidence evidence by quantile gating, pools the gated evidence across windows via Σ to compute a shared bucket support, and adds a bucket-wise bias to the original logits. In this example, the fourth window incorrectly ranks *fountain* first before GCB, but *mountain* becomes top-ranked after the shared bucket support is added.